

Kevin Worsley

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Skills

Programming Languages: C, C++, C#, Python, MATLAB, Bash scripting, familiarity with PHP, Javascript, Lua

Technologies: Git, OpenCV, Linux, Electrical + PCB Design/Layout, CAD modeling, Fusion360, EAGLE, 3D printing, Laser cutter

Professional Experience

Beaudry Interactive

North Hollywood, CA

Lead Embedded Systems Engineer

June 2024 – Present

- Directed and executed embedded hardware/PCB, software designs, and manufacturing for multiple museum exhibits
- Performed research and development to integrate proprietary Javascript application into Unity engine application
- Refined and improved novel object tracking computer vision system in Python, including on-site tuning and development
- Designed and manufactured physical prototypes for variety of interactive systems, primarily with FDM printer and laser cutter
- Performed multiple on-site installations of technical equipment, directly collaborated with local installation teams
- Collaborated closely with small multidisciplinary team to execute large-scope projects on tight timelines

Creative Technologist

October 2022 – June 2024

- Developed firmware in C/C++ to support wide variety of embedded microcontroller applications for interactive exhibits
- Researched, designed, implemented, and installed novel object tracking computer vision system utilizing OpenCV in Python
- Implemented commercial and custom sensing technology, for optimal functionality and immersion in experience design
- Oversaw and executed the electrical design, PCB layout, and manufacture of portable embedded system, 200 units

Northrop Grumman

Woodland Hills, CA

Associate Software Engineer, Test Equipment Software

February 2022 – October 2022

- Developed test files and appropriate support infrastructure in C# to validate and test aircraft navigation equipment
- Designed and implemented improvements to C# common test software architecture using object-oriented design
- Performed cross-team research to identify frequent changes in testing requirements, adapted existing software from results

Associate Software Engineer, Embedded Software

March 2021 – February 2022

- Developed logic using Simulink model-based engineering to process aircraft message buses, IO components
- Developed and maintained low-level software in C to bridge core FPGA and model environment

Technical Intern, Software

May 2020 – August 2020

- Maintained and designed Simulink models to handle messages transmitted between product and target platform
- Implemented low level bit logic to process messages in accordance with military communications standard

Education

California State Polytechnic University, Pomona

December 2020

Bachelor of Science, Computer Engineering, 3.71 GPA

Magna Cum Laude, Kellogg Honors College

Relevant Coursework

- Data Structures and Algorithms, Microcontrollers, Artificial Intelligence, Computer Architecture, Software Engineering, Digital Design with VHDL, Control Systems, Circuit Analysis, Network Forensics, Differential Equations, Physics

Academic and Personal Projects

Senior Design Project

Pomona, CA

Animatronic with Convolutional Neural Network on Embedded Platform

August 2019– May 2020

- Developed a Convolutional Neural Network using TensorFlow / Keras, OpenCV to identify and categorize facial expressions
- Passed emotion data over data bus to PIC18F microcontroller to handle motion, light, sound control for animated figure
- CNN deployed with performance constraints on Raspberry Pi Zero utilizing TensorFlow Lite platform in Python

Engineers Without Borders, CPP Chapter

Pomona, CA

Water Health Monitor, Team Lead

August 2018 – May 2019

- Developed and built Arduino-based sensor system in C++, built sensors from scratch to control costs, power consumption
- Led team of four, mentored and taught members electronics skills based on interest and existing skillsets

Northrop Grumman Collaboration Project

Pomona, CA

Unmanned Underwater Vehicle Team Member

October 2017 – May 2018

- Assisted in design of autonomous navigation algorithm in C++, implemented OpenCV library for object recognition
- Led collaboration among multiple teams to achieve systems-level integration of components, information workflow